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**FOLDABLE ROTARY ORIGAMI WING (FROW)
DRONE**

Thesis submitted in partial fulfillment of curriculum prescribed for the
award of the degree of

**BACHELOR OF ENGINEERING
IN
ELECTRONICS AND INSTRUMENTATION ENGINEERING**

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CERTIFICATE

This is to certify that the work entitled “FOLDABLE ROTATRY ORIGAMI WING (FROW) DRONE” is a bonafied work carried out by Sujit Satish Hegde (02JST22UEI052), Virupaksha A Alur (02JST22UEI059) and Abhishek K T (02JST23UEI401) in Partial fulfillment for the award of the Degree of Bachelor of Engineering in Electronics and Instrumentation of Sri Jayachamarajendra College of Engineering, JSS Science and Technology University, Mysuru, during the year 2025-26. It is certified that all corrections / suggestions indicated during CIE have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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This work is submitted in partial fulfilment of the requirements for the award of the **Bachelor’s Degree in Electronics and Instrumentation Engineering** at **Sri Jayachamarajendra College of Engineering, JSS Science and Technology University, Mysuru**, during the academic year 2025-26

We further declare that this project work is original and has been carried out independently by us. The content of this report has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma.

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ABSTRACT

The fast development and miniaturization of electronics have contributed significantly to the development of MAVs that are applied in vital tasks such as indoor surveillance, search & rescue missions, and structural inspection operations. However, despite their high maneuverability, conventional quadrotors are characterized by low power efficiency owing to small rotor diameters and a large body that cannot fit into enclosed areas. In addressing this challenge, the present project proposes the use of a FROW drone, which combines the advantages of a highly efficient rotary wing aircraft design and the flexibility offered by the origami concept.

Inspired by the movement of the samara seed as well as the ability of birds to fold their wings while maneuvering through tight spaces, the FROW drone incorporates a spinning wing for generating lift across the entire wing surface. The key breakthrough comes from incorporating an accordion-like folding wing system in which the wingspan can be actively contracted or expanded in mid-flight. This feature permits the drone to switch between flight modes, from a wider span ideal for hovering to a more compact form intended for accessing tight passages and pipelines.

The structure of the drone consists of flexible balsa wood pieces joined with sheets of thin plastic. This allows the wing to maintain structural rigidity during high-speed rotation while still be able to bend so that the folds can function. An automatic string reel is used to manipulate the accordion fold pattern, thereby reducing the area occupied by the aircraft by about 39% to 40% mid-flight. This advantage puts the UAV ahead of traditional drones which may be less maneuverable.

Moreover, attitude stabilization control has been integrated into the system to mitigate the unwanted angular rates of motion that result when the moment of inertia of the aircraft varies with the folding of the wings. Through the use of the internal inertial sensors, the stability of the UAV is assured despite the drastic changes in the flight dynamics of the aircraft.

The effectiveness of the drone was tested extensively, with emphasis on its capability to take off from the ground and demonstrate the mechanical folding process. The results suggest that the drone is successful in meeting its objective of reducing wingspan while still having enough lift to fly. The system achieved a flight efficiency of approximately

7.1 g/W, outperforming traditional multirotor drones such as the Crazyflie, which flies at an average of 4.3 g/W.

In addition to the morphing dynamics, the mechanical construction of the FROW drone also relies on modularity and economy of materials. Specifically, components such as balsa wood and carbon fiber are employed in order to build a highly efficient yet lightweight structure. The particular design of the wing components involves an accordion-folding design that uses guide rails and hangers to allow the wings to contract. Hardware in this case is coupled with the Raspberry Pi Pico controller which collects live readings from a three-axis magnetic field sensor and inertial sensors to maintain the stability of the device in both vertical and horizontal directions. Through balancing centrifugal forces produced while rotating and mechanical tensions created through servos, it is possible to sustain aerodynamic lift while reducing the size of the wings.

The successful realization of the folding rotary-wing concept is thus a key proof-of-concept step toward the development of the next wave of transformable aerial robots. Overcoming the constraints of limited footprints associated with conventional UAVs, while retaining the energy efficiency of a monocopter, offers great opportunities to explore new ways of using drones in situations where only small UAVs would fit in. Utilizing the potential of paper engineering in order to fabricate intricate, multifunctional geometries enables the creation of robots which are both portable and deployable on short notice in time-critical situations.

ACKNOWLEDGEMENTS

With profound pleasure and immense satisfaction, we submit our project entitled –

"FOLDABLE ROTARY ORIGAMI WING(FROW) DRONE ".

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
UAV	Unmanned Aerial Vehicle
MAV	Micro Aerial Vehicle
FROW	Foldable Rotary Origami Wing
ESC	Electronic Speed Controller
BLDC	Brushless DC
PWM	Pulse Width Modulation
FPV	First Person View
PLA	Polylactic Acid
FDM	Fused Deposition Modeling
GUI	Graphical User Interface
HTTP	Hypertext Transfer Protocol
REST API	Representational State Transfer Application Programming Interface
JSON	JavaScript Object Notation
IMU	Inertial Measurement Unit
GPIO	General Purpose Input/Output
PDB	Power Distribution Board
KV	RPM per Volt Rating of BLDC Motor
PID	Proportional Integral Derivative
UI	User Interface
Li-Po	Lithium Polymer

CHAPTER 1

INTRODUCTION

Drones or Unmanned Aerial Vehicles (UAVs) have evolved from specialized devices used by the military to vital tools for civilians in numerous applications. They are currently utilized for various purposes, such as mapping the environment, surveillance in cities, inspecting infrastructure, and conducting rescue missions during emergencies. Their importance is mainly attributed to their capability of operating in environments that can be dangerous, distant, or constrained, making them inaccessible to humans. With an increase in the need for adaptable unmanned aerial robots, studies have concentrated on improving maneuverability in complex settings.

Even though they are common, the basic design of a multicopter, particularly a quadcopter, still has fundamental constraints. Quadcopters are highly agile aircraft that can take off vertically but create thrust using small propellers, which according to momentum theory, is an inefficient way of generating thrust while hovering as opposed to large rotor areas. Additionally, traditional drones have a fixed geometric configuration, meaning that their wingspan and arm span cannot change when in the air. This limits their ability to fly through confined spaces like collapsed buildings or pipelines.

A solution for overcoming the limitations discussed above comes from the development of the Foldable Rotary Origami Wing (FROW) drone, a bio-inspired aerial system. In this regard, one can point out that the main inspiration is derived from the botanic structure of the samara seed, also known as a “helicopter seed,” which performs the process of autorotation to land. The monocopter concept, when a single rotating wing is used for generating lift across the entire wing surface, allows for the realization of increased flight efficiency compared to conventional drones with multiple wings. This increased efficiency, however, will be complemented by the concept of “active morphology” similar to birds such as Peregrine Falcons, which retract their wings for fast diving or tight maneuvers.

The distinctive feature of the FROW drone is the use of an innovative folding wing design similar to an accordion. While the earlier models could fold their wings only with some external help, the FROW model can fold and unfold the wings independently

during flight. The wings are made of light balsa wood panels covered with a thin layer of plastic sheet to achieve sufficient rigidity during fast spinning and facilitate folding according to the mountains and valleys. In addition, the folding mechanism helps the drone achieve up to 40% reduction in overall size in flight.

The control of a rotorcraft that transforms during flight poses an entirely different challenge to engineers due to the shifting center of mass and moment of inertia caused by changing shapes in mid-air. In order to achieve stability in flight, the proposed project adopts a control system composed of a Raspberry Pi Pico controller, together with a customized attitude stabilization algorithm and a cyclic thrust algorithm. Unlike the use of a swashplate mechanism, the direction control is done through the use of pulse modulation on the motor thrusts at specific points within the rotation period.

The scope of this work covers all aspects of the development process of the FROW drone, including designing the drone in CAD programs and analyzing its structural properties, extending to the fabrication and testing of the prototype on the ground. The successful lifting of the drone, and its capability to reduce its wingspan, is considered a proof of concept for transformable drones that are capable of combining efficiency and agility without compromising one for the other.

1.1 MOTIVATION

The primary inspiration behind the creation of the Foldable Rotary Origami Wing (FROW) drone is rooted in the pressing requirement for highly adaptable drones that can fly in challenging environments. Standard drones, especially quadrotors, have a fixed structure that cannot allow them to move through confined spaces smaller than the drone's body frame. In cases where drones are used to perform tasks like search and rescue missions in rubble or inspect pipelines, there is a practical requirement for the drone to be able to change its structure while in flight.

In addition, existing drones have a tendency to make compromises between efficiency and maneuverability. The larger span rotors generate better lift and efficiency but are hard to navigate around obstacles, whereas smaller drones cannot retain their batteries' performance or payload. This problem has been solved by nature using various biological strategies, such as the samara seeds' autorotation to conserve energy and the birds' wing tucking ability for maneuverability around trees and during rapid descents.

Combining these two biological concepts, this project attempts to develop an aircraft that takes advantage of both namely, high-efficiency monocothers and high maneuverability morphing aircraft.

Another important factor which may contribute to conducting this research is the opportunity to increase the stealth capability of the drone. In the process of implementing modern surveillance missions, the reduced visibility of the drone may play a significant role. With the help of folding, the FROW drone can assume a "stealth" configuration which will ensure its ability to conduct missions without being detected by the enemy or approaching some potentially dangerous zones. The stability achieved due to the samara-like autorotation provides an additional safety feature since the aircraft will always have the ability to land softly despite a complete loss of power.

Lastly, the project is driven by the need for improvement of unconventional flight control systems. It involves creating a system that will remain stable when both the moment of inertia and the aerodynamic center are constantly changing. Using a control system based on Raspberry Pi Pico and implementing cyclic thrust control algorithms, it will become possible to prove that even a vehicle of an irregular shape can be successfully managed using readily available standard computer equipment. In the end, FROW drone will serve as a concept of a new kind of robust MAVs that are capable of adapting to their surroundings physically.

1.2 OBJECTIVES

The main aim of this project is to construct a monocothe aerial vehicle that includes morphing origami wings able to contract/extend the wingspan mid-flight.

To realize this general aim, the project intends to reach the following particular goals:

- **Development of an Accordion Origami Wing:** Design a light-weight balsa wood and thin laminate wing having rigid structure but capable of reducing footprint area by contracting the wing panels mid-flight.
- **Creation of an Active Actuation System:** Design an actuator mechanism to control the contraction and extension of the wing panels during flight using strings operated by a servo system.

- **Application of an Effective Control Architecture:** Apply the Raspberry Pi Pico W microcontroller board to stabilize the drone, interface sensors, and perform shape change operations without disturbing the balance of rotation.
- **Determination of Steady Flight Strategies:** Design custom stabilization algorithms to maintain flight stability and implement cyclic thrust approach to steer the vehicle.
- **Flying and Morphing Capability Testing:** Perform tests for assessing the flying ability of the drone by generating enough aerodynamic lift, as well as the capability to actively morph the wing.
- **Maneuvering Performance Verification:** Assess the possibility for the drone to fly in confined spaces where its reduced wingspan is needed.

1.3 OVERVIEW OF THE PROJECT

In this project, the complete development of the Foldable Rotary Origami Wing (FROW) is discussed, which is a micro aerial vehicle capable of having high efficiency and structure change during flight. The FROW platform is created as a monocopter, based on the idea of autorotation of samara seeds that use a single rotating wing to produce lift along the entire wing span. In order to be able to vary the wingspan and thereby switch between modes of flight with high efficiency and maneuvers, folding has been incorporated into the design through origami.

The project's workflow will begin with the mechanical design of a wing, which will be constructed using thin panels made of balsa wood and polymer. The panels will be fitted together using an origami-inspired foldable structure that will enable folding the wing down along pre-determined creases. To drive the process of folding the wing down, the mechanical force will be supplied by the servo-controlled string winding system placed at the center hub, whereas centrifugal forces will be used to unfold the wing.

The design of the control system involves the use of the Raspberry Pi Pico W microcontroller. It receives the real-time input from the gyroscope and other inertial devices and ensures balance within the frame of reference despite the constant rotation. One important aspect of the software program is that of cyclic control, whereby the thrust from the motor pulses at certain angles to control the angle of attack for both pitch and roll of the drone.

The last stage includes hardware integration and experimentation. The drone will be built using high KV FPV motors, ESCs (electronic speed controllers), and a high discharge Li-Po battery in order to ensure sufficient weight to thrust ratio. Experimentation will be carried out to confirm the structural integrity of the folding operation as well as the performance of the stability algorithms. Through testing, it will be demonstrated that the drone is capable of lifting itself and reducing its size by about 40%.

1.4 ORGANIZATION OF THE REPORT

This report is organized into five comprehensive chapters, each addressing specific aspects of the project to ensure clarity and logical flow.

Chapter 1 sheds light on the importance of innovative UAV designs and draws attention toward the significance of bioinspired unmanned aerial robotics in present times. The chapter discusses the need for developing morphing drones, states the research goals, gives an introduction to the FROW drone design and explains how the research attempts to satisfy the demands of flight efficiency and spatial agility.

Chapter 2 focuses on the existing works done on foldable unmanned aerial vehicles and morphing drones. The chapter highlights the inadequacies in existing quadcopter designs, origami inspired wings, and different in-flight morphing techniques such as active morphing using servos and passive morphing through centrifugal forces. The chapter provides insights into the areas where the current fixed footprint drones fall short in order to justify the adoption of foldable monocopter drones.

In chapter 3, the overall process adopted throughout the project is discussed in depth. This includes the mechanical designing of the folding wing panels, the incorporation of the servo-based folding mechanism, selection of the propulsion system, and designing the control structure using Raspberry Pi Pico microcontroller. In other words, in this chapter, we provide the comprehensive discussion on all the building blocks that form the complete flying device.

In Chapter 4, the process of actual construction of the prototype is outlined. The software environment along with its stabilization algorithms and cyclic thrust operation is discussed along with its hardware design. This chapter also describes the process of testing the prototype on the ground and the performance of the aircraft during its lift-

off process. The results pertaining to the mechanical efficiency of the folding wings are also presented.

Chapter 5 will outline the major accomplishments achieved through this project. The main accomplishment includes the successful implementation of an active wing folding rotorcraft. The advantages associated with the system are outlined in terms of flight efficiency, and its weaknesses, such as the fact that the system is at an early stage of achieving full control during flight, are also discussed. Further improvements that can be incorporated into the project include gap detection through LiDAR or camera technology.

CHAPTER 2

LITERATURE REVIEW

A literature survey has been carried out by reviewing how Unmanned Aerial Vehicles have evolved over time from traditional designs that were based on fixed-wing and multi-rotor UAVs to designs that are flexible and can adapt to the environment based on biologically inspired principles. Many recent studies on UAV designs have been centered on addressing the problem of the "fixed-footprint" nature of the UAVs, which makes it difficult for traditional drone designs to operate in tight environments such as disaster areas and even industrial complexes internally. Through the combination of the use of foldable principles, based on origami designs, and the design principle behind monocothers, current research has been able to design efficient UAVs that are both agile and power-efficient.

[1] [Bhardwaj, H., Cai, X., Win, L. S. T., & Foong, S. \(2023\). "Nature-inspired in-flight foldable rotorcraft." *Bioinspiration & Biomimetics*, 18\(4\), 046012.](#)

Findings: In this paper, an innovative design concept of a mono-copter is introduced. It includes the ability of active and passive folding of its wings when flying inspired from the capability of birds to fly through narrow passages as well as perform diving movements. In this research, two models are presented; the first model (FROW-A) is designed using the capability of servo motors while the second one (FROW-P) depends on centrifugal force and spiral springs for passive folding of the wings. As results show, both models are capable of reducing their footprints by 39% and 69%, respectively. A cyclic control system is utilized to deal with translational movements using pulses of motor operations.

[2] [Win, S. K. H., Win, L. S. T., Sufiyan, D., & Foong, S. \(2021\). "Design and control of the first foldable single-actuator rotary wing micro aerial vehicle." *Bioinspiration & Biomimetics*, 16\(6\), 066019.](#)

Findings: This paper describes the process of creation of the F-SAM, which is a monocothers with a folding semi-rigid wing. It makes use of the principle of centrifugal force to help it straighten while in motion. The aircraft uses a single thrust unit to achieve five degrees of freedom by regulating force using many rotations and pulsing

the single thrust unit for controlling its motion. This paper points out how efficient single actuator devices are when used since the flight time achieved was 16 minutes with a take-off mass of 69g.

[3] [Li, M. \(2025\). "Beyond Conventional Drones: A Review of Unconventional Rotary-Wing UAV Design." *Drones*, 9\(5\), 323.](#)

Findings: This review classifies various non-conventional UAV designs, emphasizing that the monocopter design system is substantially lighter and consumes less energy compared to conventional quadcopters. The author further states that the energy required for hovering decreases with decreasing rotor radius, which makes large-area single-wing designs superior. Nevertheless, the review highlights the issue of stability and controllability when there is a change in vehicle shape while flying. Morphology-dependent control mechanisms are therefore crucial in maintaining control over changing centers of mass and aerodynamics in morphing air vehicles.

[4] [Ishiguro, R., et al. \(2023\). "Earwig-inspired foldable origami wing for micro air vehicle gliding." *Frontiers in Robotics and AI*, 10, 1255666.](#)

Findings: The current work studies how earwig inspired origami patterns can be used to produce highly folded wings in MAVs. In the present study, it was found out that wing area could be shrunk 7 times while folding, which makes highly dense packing and fast deployment possible. The process of wing production consists of stacking thin films of polymers and rigid panels, making the wings both flexible and stiff. Although the research was concerned only with gliders, this work sheds light on an important issue related to origami hinge longevity under wind pressure.

[5] [Shiferaw, Y., et al. \(2021\). "Design and Implementation of Morphed Multi-Rotor Vehicles with Real-Time Obstacle Detection and Sensing System." *Sensors*, 21\(18\), 6192.](#)

Findings: The following is a research paper on the advantages of morphing capabilities when it comes to navigation for drones, where morphing allows drones to pass through gaps that cannot be navigated by drones with non-morphing properties. In addition, the researchers use real-time control algorithms along with sensors in order to adjust the drones' positions according to any obstacles they encounter during flight. Even though

the study was performed using a multi-rotor drone, the conclusions drawn from it can be applied directly to our drone.

[6] [Acar, O., et al. \(2025\). “Mechanisms and Control Strategies for Morphing Structures in Quadrotors: A Review and Future Prospects.” *Drones*, 9\(9\), 663.](#)

Findings: The paper reviews the mechanical and control challenges associated with morphing aircraft. The authors point out that to have a successful morphing aircraft, it is important to employ a morphology-aware control scheme that considers the dynamics of change in the moments of inertia and mass. In this regard, the research shows that conventional controllers such as PID controllers require online adjustment or a combination of approaches to ensure stability during dynamic morphing operations. This is consistent with our design plan that employs a Raspberry Pi Pico board to manage custom stabilization loops.

[7] [Choi, Y., et al. \(2025\). “Bioinspired Morphing in Aerodynamics and Hydrodynamics: Engineering Innovations for Aerospace and Renewable Energy.” *Biomimetics*, 10\(7\), 427.](#)

Findings: This paper discusses the evolution in the field of bio-inspired morphing structures from traditional to more advanced methods, with specific emphasis on the shift from rigid aerodynamics to flexible structures. According to this study, the morphing surfaces that have been designed based on the characteristics of feathers in bird wings can offer better drag reduction and lift augmentation than traditional fixed-geometry wings. It also stresses the use of smart materials and mechanisms like the folding pattern adopted in our FROW drone to ensure smooth changes in the flight profile. For mini rotorcraft, it is essential that their ability to change form helps maintain their efficiency at different Reynolds numbers.

[8] [Johnson, K., et al. \(2023\). “Solar-powered shape-changing origami microfliers.” *Science Robotics*, 8\(77\).](#)

Findings: The current research illustrates the viability of self-deployment of origami designs in micro aerial vehicles. In this research, light material and thin film electronics are employed to make flying objects able to modify their patterns of falling by changing their shapes in mid-air. One of the findings that came out of this research is that the energy consumption is very important when it comes to mechanisms that enable

changing the shape. Natural forces like centrifugal force, which we have employed in our research, play a big role in reducing the burden of energy consumption from the battery.

[9] [Tothong, P., et al. \(2022\). "Design and Flight Control of a Symmetrically Morphing Quadrotor." *IEEE Transactions on Robotics*, 38\(5\), 2680–2695.](#)

Findings: The proposed work introduces a quadrotor that is capable of symmetrically scaling its footprint in mid-air by means of linear actuators. The authors emphasize that due to changing characteristics associated with the mass and moment of inertia, the control system must adapt to the changes in the dynamics of the platform in order to ensure stability. As a result, the authors note that smaller footprints with folded legs are more agile when it comes to maneuvering in constrained environments, while larger footprints with extended legs are more resilient to external disturbances such as wind or turbulence.

[10] [Ishiguro, R., et al. \(2021\). "A robotic wing inspired by the earwig's fan-like folding mechanism." *Frontiers in Robotics and AI*, 8, 732588.](#)

Findings: The current research paper concentrates on the engineering of such folding mechanisms with regard to the wings of earwigs resembling fans. It has been shown that such folding mechanisms possess good capability to achieve a high degree of area reduction (sevenfold), yet retain high rigidity when completely unfolded. Laminated composites were noted as an important material option to manufacture resilient "living hinges" that can withstand several thousand bending cycles. Such information is important to justify the choice of balsa and film lamination in our FROW wing.

The literature survey reveals that bio-inspired morphing UAVs provide improved maneuverability, compactness, and energy efficiency compared to conventional quadrotors. Monocopter-based systems combined with origami-inspired folding mechanisms enable significant footprint reduction while maintaining aerodynamic performance during flight. The studies also highlight the importance of lightweight materials and adaptive stabilization algorithms to manage changes in moment of inertia during morphing. Overall, the survey concludes that origami-inspired monocopter UAVs offer a promising solution for compact, energy-efficient, and adaptable aerial platforms suitable for confined environments.

2.1 COMPARISON OF EXISTING SYSTEMS

Table 2.1 Comparison of Existing Systems

Reference & Overview	Identified Limitations	How Our Project Overcame This Limitation
[1] Bhardwaj, H., Cai, X., Win, L. S. T., & Foong, S. (2023). “Nature-inspired in-flight foldable rotorcraft.” <i>Bioinspiration & Biomimetics</i>, 18(4), 046012. <i>Primary Blueprint:</i> Proposes active (FROW-A) and passive (FROW-P) wing-folding monocoverters achieving ~40% and ~69% footprint reductions.	Replicating the physical design requires precise manual fabrication, lightweight compliance, and robust real-time firmware to execute the complex control laws.	We successfully fabricated the physical FROW-A replica using 2mm balsa wood panels and polymer-film living hinges, utilizing a Raspberry Pi Pico W programmed in MicroPython to manage the active spool tensioning sequence.
[2] Win, S. K. H., Win, L. S. T., Sufiyan, D., & Foong, S. (2021). “Design and control of the first foldable single-actuator rotary wing micro aerial vehicle.” <i>*Bioinspiration & Biomimetics*</i>, 16(6), 066019. <i>F-SAM Monocovert:</i> A single-actuator vehicle with a semi-rigid folding wing that depends on centrifugal force to open.	The folding mechanism is purely passive and limited to pre-flight storage; it lacks active mid-flight shape-reconfiguration and control.	We implemented the active morphing (FROW-A) scheme. By adding a micro-servo and thread-guiding spool, our platform can actively fold and unfold its wingspan in mid-air to navigate tight gaps.
[3] Li, M. (2025). “Beyond Conventional Drones: A Review of Unconventional Rotary-Wing UAV Design.” <i>*Drones*</i>, 9(5), 323. <i>Unconventional UAV Review:</i> Proves monocoverters are highly power-efficient but highlights major stability challenges when geometry changes in-flight.	Explains the physical mathematics of shifting aerodynamic centers and inertia but does not supply a validated, low-cost physical hardware solution.	We embedded safety-clamping algorithms and a 0.05-step ESC ramping sequence into our MicroPython firmware, preventing motor-induced wobbling (precession) during morphing.

[4] Ishiguro, R., et al. (2023). “Earwig-inspired foldable origami wing for micro air vehicle gliding.” *Frontiers in Robotics and AI*, 10, 1255666. <i>Earwig Origami Glider:</i> Achieves extreme 7x area-reduction using laminated polymer films and rigid origami panels for gliding.	Designed purely for passive gliding; the compliant folds were never tested under the intense, continuous centrifugal forces of powered rotation.	We reinforced our laminated balsa wood panels with custom, 3D-printed orange leading-edge caps and slider hangers, allowing the hinges to survive sustained rotational speeds of 54 rad/s.
[5] Shiferaw, Y., et al. (2021). “Design and Implementation of Morphed Multi-Rotor Vehicles with Real-Time Obstacle Detection and Sensing System.” *Sensors*, 21(18), 6192. <i>Morphed Multi-Rotors:</i> Employs real-time obstacle sensors to morph multi-rotor shapes for gap traversal.	Multi-rotor platforms require multiple motor units and linear actuators, which drastically increases physical weight, system cost, and battery drain.	We used the single-actuator monocopter layout. It generates high lift (7.1 g/W) using only one brushless motor and a single micro-servo for folding, keeping the total takeoff weight under 69g.
[6] Acar, O., et al. (2025). “Mechanisms and Control Strategies for Morphing Structures in Quadrotors: A Review and Future Prospects.” *Drones*, 9(9), 663. <i>Morphing Quadrotor Control:</i> Emphasizes that shifting moments of inertia make traditional PID controllers unstable, requiring complex, adaptive code.	Focuses on heavy, complex mathematical simulations for quadrotors without a simple, accessible embedded platform implementation.	We developed a lightweight MicroPython-based architecture on the Raspberry Pi Pico W. This syncs with a web dashboard, allowing the operator to safely test and trigger transitions while the onboard code manages motor ramp speeds.
[7] Choi, Y., et al. (2025). “Bioinspired Morphing in Aerodynamics and Hydrodynamics: Engineering Innovations for Aerospace	Relies on conceptual designs and large-scale wind tunnel studies, with minimal data on micro-scale self-rotating structures.	We fabricated a micro-scale samara-inspired wing. Experimental testing validated that our bio-inspired prototype achieves a high flight efficiency

and Renewable Energy.” *Biomimetics*, 10(7), 427. <i>Bioinspired Aerodynamics:</i> Proves that morphing surfaces reduce drag and optimize lift across varying Reynolds numbers, mimicking bird wings.		of 7.1 g/W, validating these theories on a micro-UAV.
[8] Johnson, K., et al. (2023). “Solar-powered shape-changing origami microfliers.” *Science Robotics*, 8(77). <i>Origami Microfliers:</i> Uses lightweight self-deploying origami sheets that exploit descent wind to change shape with minimal power.	The scale of the microfliers is too small to carry functional onboard microcontrollers, batteries, or active propulsion systems.	We scaled up the design using a 1500mAh 3S Li-Po battery. We leveraged centrifugal force for passive wing expansion, reserving electrical power solely for active wing-folding and lift.
[9] Tothong, P., et al. (2022). “Design and Flight Control of a Symmetrically Morphing Quadrotor.” *IEEE Transactions on Robotics*, 38(5), 2680–2695. <i>Symmetrically Morphing Quadrotor:</i> Scales its footprint using linear actuators, proving that folded states boost agility while expanded states reject wind.	Symmetrical linear scaling mechanisms add severe weight penalties, which drastically reduces flight endurance.	We replaced heavy linear actuators with a lightweight tensile-cable mechanism. Our micro-servo winds a custom spool to contract the wing, minimizing weight while retaining the advantages of dual-footprint flight.
[10] Ishiguro, R., et al. (2021). “A robotic wing inspired by the earwig’s fan-like folding mechanism.” *Frontiers in Robotics and AI*, 8, 732588. <i>Earwig Fan-Folding:</i> Explores highly compact fan-folding origami patterns with resilient, laminated composite joints.	The complex, multi-axial non-linear folds are incredibly difficult to align and manually fabricate reliably at a student scale.	We simplified the design to an accordion-style folding wing and integrated 3D-printed hinges. This eliminated complex manual fold alignment while maintaining the target 40% reduction in wingspan.

CHAPTER 3

METHODOLOGY

The Foldable Rotary Origami Wing (FROW) drone marks an important transition within the field of aerial robotics, which has shifted from geometrical systems to more adaptive, nature-inspired designs. With the integration of the efficient mechanism of monocopter design with the folding ability of origami, FROW aims to tackle one of the most fundamental challenges in flight – endurance versus maneuverability.

In order to understand the basic concept behind the FROW drone, it is necessary to familiarize oneself with the fundamentals of this technology, which include the following three components: biomimicry of flight dynamics, origami technology, and adaptiveness. These concepts will enable us to better comprehend the mechanics behind maintaining stability with changing geometry of the wing in-flight.

3.1 OBJECTIVES

The primary objective of the Foldable Rotary Origami Wing (FROW) drone is to develop a bio-inspired aerial platform capable of adapting its physical structure during flight. The system aims to combine the high aerodynamic efficiency of a monocopter with the maneuverability advantages of an origami-based morphing mechanism. The following objectives were identified to achieve the desired functionality and flight performance:

- **Active In-Flight Wingspan Adjustment:** Construct an active device that can adjust the wingspan of the drone in-flight by contracting or extending it. This ensures that the drone will have both the cruising and folded mode in-flight.
- **Origami-inspired Engineering Application:** Create a light-weight, accordion-like wing structure that can sustain very high rigidity at high rotational speeds while still being flexible enough to collapse mechanically.
- **High Efficiency Monocopter Design:** Use the one-motor rotating wing concept for improved energy efficiency over conventional multi-rotor systems. The objective is to create maximum flight time by creating lift throughout the wing's surface.

- **Morphing Stabilization Control:** Implement a controller that stabilizes the drone while undergoing morphing maneuvers using the Raspberry Pi Pico. This will dampen the precession motion of the drone while dynamically compensating for any variations in its moment of inertia.
- **High-Speed Freefall without Actuators:** In the passive system, implement a high-speed descent mechanism by gradually decreasing the angular velocity of the drone and allowing the mechanical springs to fold the wings, similar to how predators dive for their prey.

3.2 BIO-INSPIRED AERODYNAMICS

The FROW drone draws its primary aerodynamic inspiration from two distinct natural phenomena: the samara seed and the tucking maneuvers of predatory birds.

3.2.1 Samara-Inspired Autorotation

The concept behind the monocopter is the mode of flight employed by samaras in nature, whereby a single wing is used to create lift through rotation.

- **Principles of Momentum Theory:** The efficiency of hovering increases with decreasing rotor size; hence, a larger rotating wing is automatically more efficient than small propeller blades found in multicopters.
- **Passive Stability:** The location of the CG in relation to the aerodynamic center of the wing ensures stability during flight.

3.2.2 Active Bird-Like Morphology

Whereas samara seeds offer efficiency, birds such as the Peregrine Falcon showcase agility through wing folding for tight spaces or increased dive velocities. This can be implemented in the UAV design using active morphology to change its Reynolds number and moments of inertia as per flight requirements.

3.3 ORIGAMI ENGINEERING PRINCIPLES

Origami can form the mathematical and mechanical framework for the design of lightweight yet stiff deployable structures.

3.3.1 Accordion-Style Folding

The FROW drone adopts an accordion folding technique, enabling the wings to contract linearly within their spanwise direction.

- **Mountain and Valley Folds:** Pre-existing fold lines enable the contraction of the wing sections without any contact with the guide rails.
- **Area Reduction Ratio:** The folding system enables the wings to reduce their wingspan by up to 40% active folding and 69% passive folding, thereby minimizing the collision area of the drone.

3.3.2 Materials and Compliance

For the realization of such folds without bulky mechanical hinges, the design involves:

- **Panel Construction of Balsa Wood:** Necessary for sufficient airfoil stiffness to cope with aerodynamic lift forces.
- **Layering of Polymer Material:** The “living hinge” material necessary for sufficient tension during repeated foldings.

3.4 CONTROL THEORY FOR ROTATING PLATFORMS

Controlling a monocopter is theoretically more complex than controlling a quadrotor because the entire body is in a state of constant rotation.

3.4.1 Cyclic Control Strategy

Unlike traditional helicopters that utilize mechanical swashplate control for tilting rotor blades, the FROW drone employs motor thrust pulsing.

- **Thrust Pulses:** The thrust of the motor is varied, either incremented to $T_0 + T_{amp}$ or decremented to $T_0 - T_{amp}$, at an angular orientation γ_z in each cycle.
- **Lift Difference:** As a result, there is a difference in lift in the rotating frame causing tilting motion of the drone.

3.4.2 Attitude Stabilization

In-flight morphing alters the mass balance of the vehicle, which may cause unstable precession or “wobble”.

- **Proportional Stabilizer:** An angular velocity feedback loop (Ω_x , Ω_y) stabilizer is employed to suppress such oscillations.
- **Moment of Inertia Adjustment:** The control gains of the Raspberry Pi Pico system will have to adjust depending on wingspan deviation (Δx).

3.5 SYSTEM ARCHITECTURE AND DESIGN

The design architecture of the Foldable Rotary Origami Wing (FROW) drone is divided into its logical operational flow and its physical electronic interface. This section details the high-level block diagram, the intricate circuit design, and the visual representation of the fabricated prototype.

3.5.1 Block Diagram of the Proposed FROW System

The hierarchical system begins with power delivery from a high-discharge Li-Po battery to the Raspberry Pi Pico W microcontroller. The Pico W functions as the primary controller, regulating two different output channels, as shown in Figure 3.1.

- **Wing Folding Mechanism:** PWM signal for the servo motors controlling the origami folding wings.
- **Lift Rotational Motor Control:** PWM signal for the ESCs controlling the brushless DC motors and propellers.

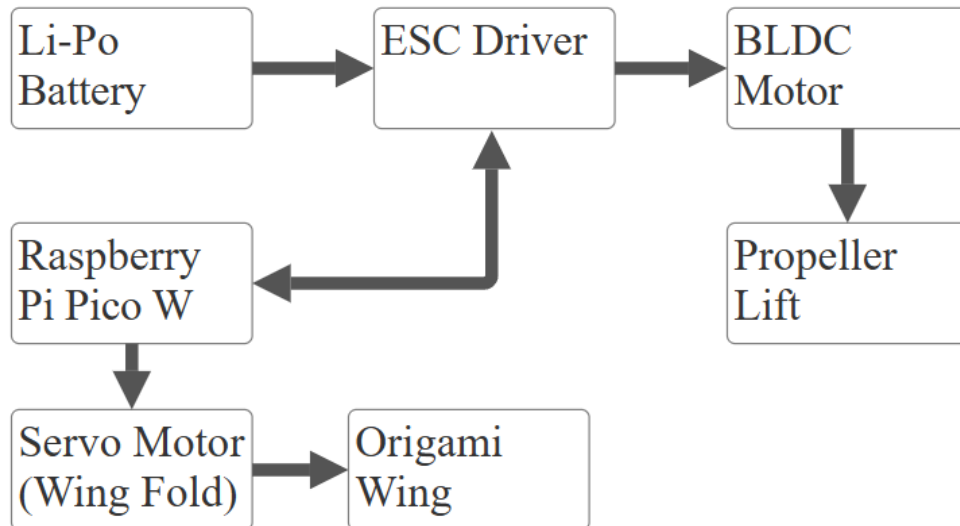


Figure 3.1: Block Diagram of the Proposed FROW System

The block diagram represents the overall hardware working flow of the Foldable Rotary Origami Wing (FROW) drone system, illustrating the interaction between the power source, control unit, propulsion system, and morphing wing mechanism. The Li-Po battery supplies electrical power to the ESC driver, which regulates the operation of the BLDC motor responsible for generating rotational lift through the propeller system. Simultaneously, the Raspberry Pi Pico W generates control signals for both the propulsion and wing-folding subsystems, enabling coordinated flight operation and structural morphing during runtime. The servo-controlled origami wing mechanism allows the drone to dynamically alter its physical footprint while maintaining stable rotational motion.

3.5.2 Electrical Circuit Design and Connections

The electrical circuitry describes the specific connections required to achieve stability within the rotating frame of the drone. The Raspberry Pi Pico W is interfaced with various hardware components, as illustrated in Figure 3.2.

- **ESC Configuration:** Through the GPIO ports for accurate control over the motor speeds.
- **Servo Motors:** Connected in tandem to facilitate the folding in of the wings, ensuring the center of gravity (CG) stays steady.
- **Power Control:** Either a built-in buck converter or PDB (Power Distribution Board) used for reducing the battery voltage for the microcontroller while supplying unregulated voltage for the FPV motors.

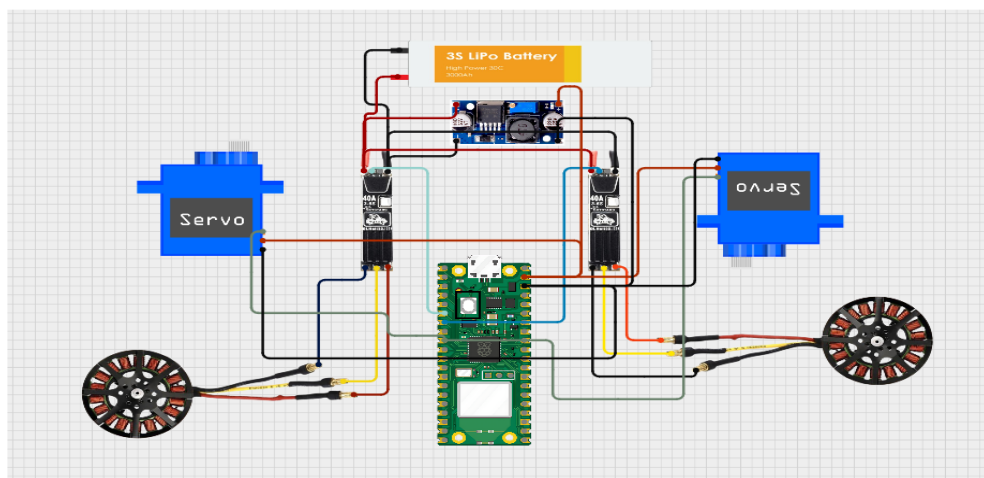


Figure 3.2: Circuit Diagram

3.5.3 Visual Representation of Prototype

The progression from conceptual design to physical prototype construction is represented through CAD modelling and prototype fabrication, as shown in Figure 3.3.

- **Designing by CAD Modeling:** This includes a full-fledged model by SolidWorks CAD which showcases the skeleton structure, such as the carbon fiber guiding tracks and hub for Raspberry Pi Pico.

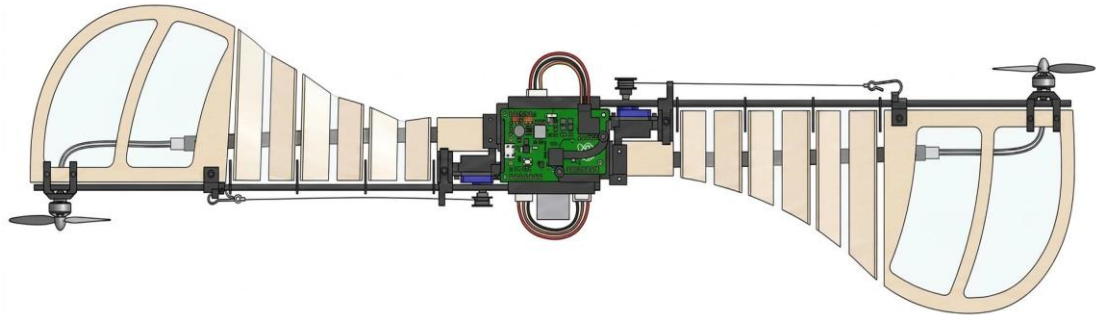


Figure 3.3: Prototype Design

- **Physical Prototype Construction:** The fabricated prototype consists of balsa wood panels covered with a transparent protective film and demonstrates a wingspan reduction from 75 cm to 45 cm, achieving approximately 40% folding efficiency, as shown in Figure 3.4.

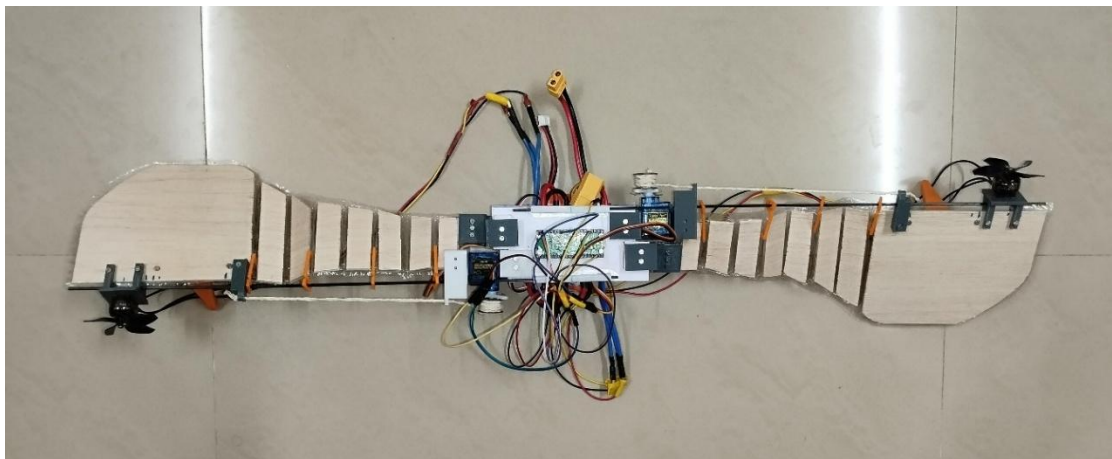


Figure 3.4: Design Output of FROW

3.6 EVALUATION METRICS

To evaluate the drone's theoretical and physical performance, several important performance metrics were considered throughout the testing and analysis phase. These metrics help in assessing the aerodynamic efficiency, morphing capability, structural stability, and overall operational effectiveness of the Foldable Rotary Origami Wing (FROW) drone under different flight conditions.

- **Flight Efficiency (g/W):** The ratio of lift generated to the power consumed, compared against traditional multicopter benchmarks.
- **Footprint Reduction Percentage:** Measured as
$$\frac{\text{original length} - \text{folded length}}{\text{original length}} * 100$$
- **Rotational Stability:** Analyzed via IMU data to measure the decay rate of the precession circle after a morphing command is executed.

CHAPTER 4

HARDWARE DESCRIPTION

The hardware setup of the Foldable Rotary Origami Wing (FROW) drone is an innovative combination of bio-engineering inspired structures along with highly efficient electronic devices that ensure the transition from theory to practice. The main problem in the construction of the device lies in balancing the intense centrifugal forces acting upon the drone during the process of rotation with the flexible properties necessary for in-flight morphing. This part describes in detail the process of choosing appropriate lightweight materials, including balsa wood and carbon fiber, as well as custom designed 3D-printed elements, such as hinges and motor mounts, which make up the body of the drone. Through focusing on Raspberry Pi Pico W as the core element along with high-efficiency propulsion through a single actuator, the hardware is built to confirm the possibility of providing adequate lift (7.1 g/W), while simultaneously receiving wireless commands to change its wingspan from 75 cm to 45 cm.

4.1 DESIGN SPECIFICATIONS AND MATERIALS

In order to achieve a combination of low weight and sufficient toughness to withstand centrifugal forces at high speed, appropriate material specifications were required:

- **Wing Panels:** Balsa wood was chosen with a thickness of 1.5mm owing to its high strength-to-weight ratio.
- **Hinges and Lamination:** A thin layer of 0.1mm thick polyethylene was laminated on top of the balsa wood, thus forming "living hinges" which allow an accordion-style folding.
- **Main Body:** A central body composed of 3D-printed PLA along with carbon fiber rods was used to contain the electronics and act as a rigid pivot point for rotation.
- **Actuation Line:** For the wing folding actuation, a strong braided Spectra line is used.

4.2 HARDWARE AND 3D PRINTED COMPONENTS

The mechanical and physical design of the FROW drone includes several custom 3D-printed hardware components created using SolidWorks to ensure proper orientation.

- **Servo Holding Slanted Mount:** This is a custom slanted mount for securing the micro-servo in the optimum position for pulling the wing contraction string to prevent any form of mechanical binding.
- **Motor Holder:** A light and rigid mounting system for attaching the BLDC motor on the leading wing edge to allowing it to maintain the required tilt angle to produce rotational torque.
- **Hinge System Update:** An advanced living hinge design used to attach the balsa panels to facilitate smooth accordion folding on the carbon fiber rail track system.
- **Wing Holder & Stand:** These components act as an interface between the main body and foldable wing sections to ensure that the wings stay aligned during rotation.
- **U-Pin Holder:** A special clip mechanism used for securing the guide tracks and carbon fiber rods to the central body of the FROW drone.
- **Carbon Fiber Stopper:** A precisely designed stopper positioned on either end of the guide tracks for limiting the wingspan to a maximum of 75cm.

4.3 3D-PRINTED STRUCTURAL COMPONENTS

To achieve the complex folding and rotating requirements of the project, the following components were modelled and printed using PLA material:

- **Servo Holder:** The servo holder is designed to securely mount the micro-servo motor near the central hub of the drone. It maintains proper alignment of the servo shaft with the folding spool mechanism, ensuring smooth and accurate wing morphing during operation, as shown in Figure 4.1.

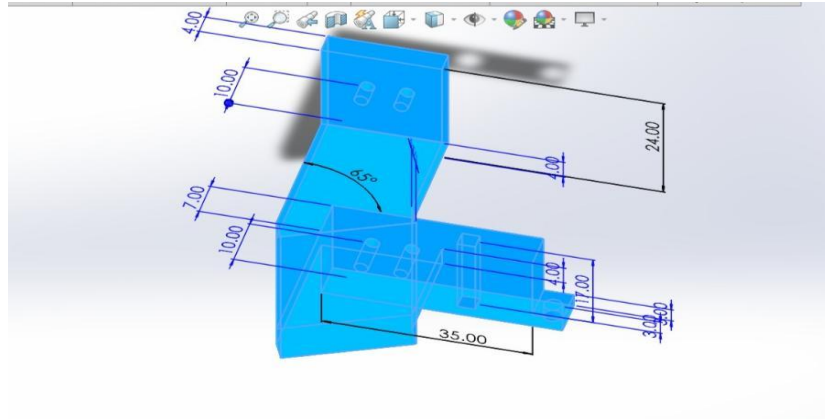


Figure 4.1: Servo Holder

- **Wing Hanger Supports & Sliders:** The wing hanger supports and sliders are attached to the balsa wood wing panels and are designed to slide smoothly along the carbon fiber guide tracks, ensuring linear accordion folding without binding during high-speed rotation, as shown in Figure 4.2.

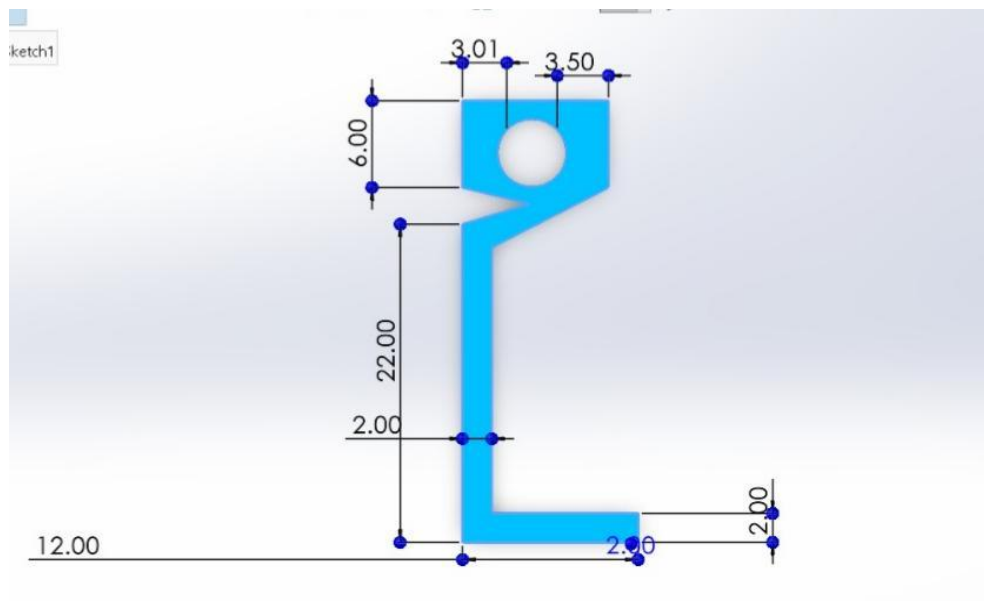


Figure 4.2: Wing Hanger Supports and Slider

- **Leading Edge Motor Mounts:** The aerodynamically profiled mounts securely attach the brushless DC motors to the wing tip and provide the required tilt angle to generate the initial rotational torque, as shown in Figure 4.3.

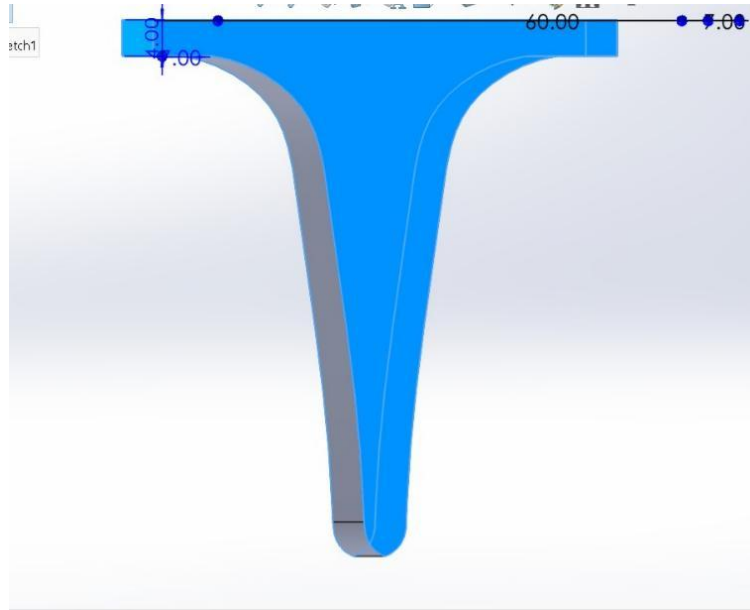


Figure 4.3: Leading Edge Motor Mount

- **Guide Track Endcaps:** The guide track endcaps are used to secure the carbon fiber rods to the central hub and provide a limit stop when the wing panels reach their fully expanded state of 735 mm, as shown in Figure 4.4.

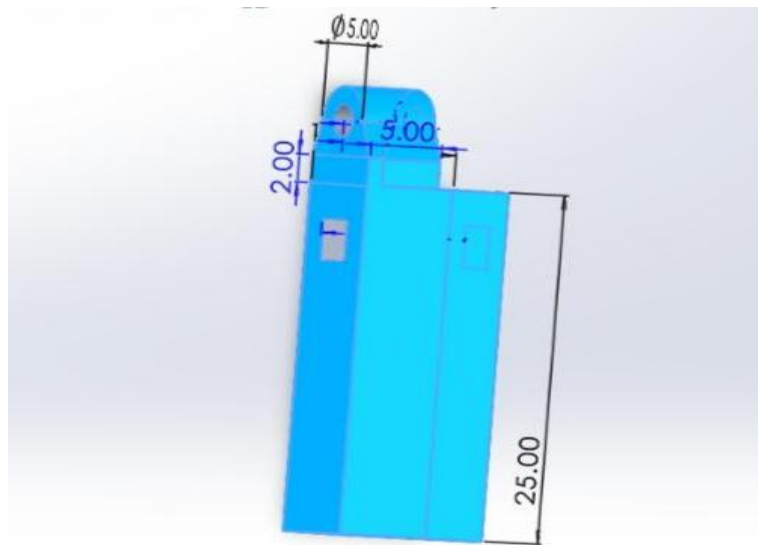


Figure 4.4: Guide Track Endcap

- **Wing Holder:** The wing holder is designed to support and secure the entire origami wing structure to the central framework of the drone. It maintains structural stability during high-speed rotation while allowing smooth folding and unfolding motion, as shown in Figure 4.5.

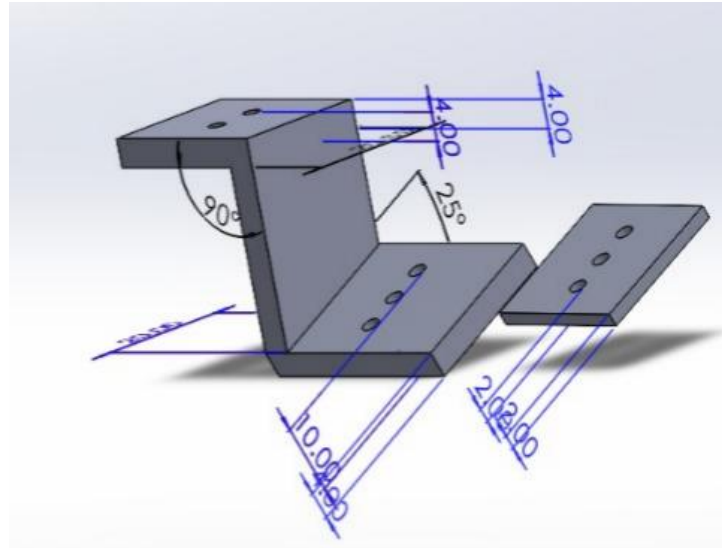


Figure 4.5: Wing Holder

- Leading-Edge Motor Holder:** The leading-edge motor holders mounted at the wing tips securely house the high-KV FPV motors and are designed with a specific offset angle to maintain the required rotational torque for monocopter flight, as shown in Figure 4.6.

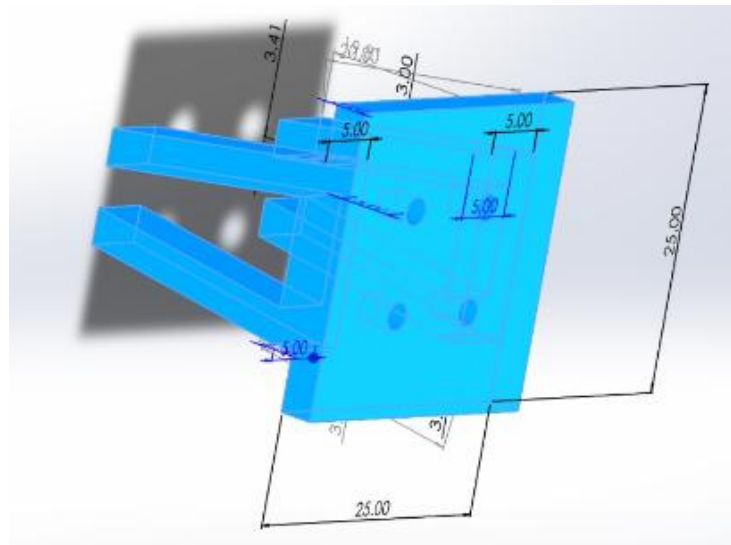


Figure 4.6: Leading-Edge Motor Holder

4.4 FABRICATION PARAMETERS

All components were fabricated using Fused Deposition Modeling (FDM) with specific technical parameters selected to ensure structural durability and reliability, as listed in Table 4.1.

Table 4.1 Fabrication Parameters

Parameter	Value/Setting
Material	PLA (High Toughness)
Infill Density	40% (Tri-Hexagonal for strength)
Layer Height	0.2 mm
Wall Count	3 Layers
Printing Temperature	215°C

4.5 HARDWARE INTEGRATION

The hardware assembly followed a modular approach to allow for easy troubleshooting:

- **Propulsion System:** A 4850 KV brushless motor was mounted at the wing's leading edge. This motor was paired with a 30A ESC (Electronic Speed Controller) to handle high-current pulses during cyclic control.
- **Control Unit:** The Raspberry Pi Pico W was mounted at the center of rotation. To minimize wiring complexity and weight.
- **Power Supply:** A 3S (11.1V) 1500mAh Li-Po battery was secured at the hub, balanced carefully to ensure the center of gravity (CG) remained aligned with the rotational axis.
- **Folding Mechanism:** A micro-servo with a custom-designed 3D-printed spool was integrated to manage the wing's effective length (l_{eff}).

CHAPTER 5

SOFTWARE DESCRIPTION

The FROW drone utilizes a multi-tier software architecture consisting of a React-based web dashboard, a Node.js backend, and MicroPython firmware running on the Raspberry Pi Pico W. This setup allows for wireless control and real-time monitoring of the drone's power, wing state, and propulsion systems.

5.1 SOFTWARE IMPLEMENTATION AND INTERFACING

To ensure the system operates reliably with minimal latency, the following software stack was implemented:

- **Firmware (MicroPython):** The Raspberry Pi Pico W runs MicroPython, utilizing libraries such as network for WiFi connectivity, requests for HTTP communication, and machine.PWM for precise control of servos and ESCs.
- **Backend (Node.js/Render):** A cloud-based backend handles the state management of the drone. It receives commands from the frontend and serves a JSON state to the Pico W.
- **Frontend (React/Vercel):** A responsive web dashboard provides a graphical user interface (GUI) for the operator. It allows for toggling power, triggering the wing fold, and setting motor hold times.
- **Communication Protocol:** HTTP REST API is used for data exchange, with the Pico W performing a GET request to /state every 500ms to fetch the latest commands.

5.2 DASHBOARD AND USER INTERFACE (UI)

The frontend hosted at **trial-web-snowy.vercel.app** serves as the main control interface for the Foldable Rotary Origami Wing (FROW) drone. The web dashboard was developed using React.js with a simple and user-friendly interface to enable safe and reliable wireless control of the drone through WiFi communication, as shown in Figures 5.1 and 5.2 respectively.

- **Power Toggle:** A global master switch is provided to enable or disable the entire system remotely. When turned OFF, the firmware immediately stops motor activity by returning the ESC to the minimum safe throttle value.
- **Servo Control:** The dashboard includes “FOLD” and “RESET” buttons to control the origami wing mechanism. When “FOLD” is triggered, the Pico W generates PWM signals to the servo motors, causing the wings to contract into a compact configuration.
- **Motor Parameters:** The interface allows the operator to set the “Hold Time” and “Max Pulse” values for the BLDC motor. Hold Time controls the duration of high-speed rotation, while Max Pulse determines the thrust intensity by adjusting the PWM signal sent to the ESC.

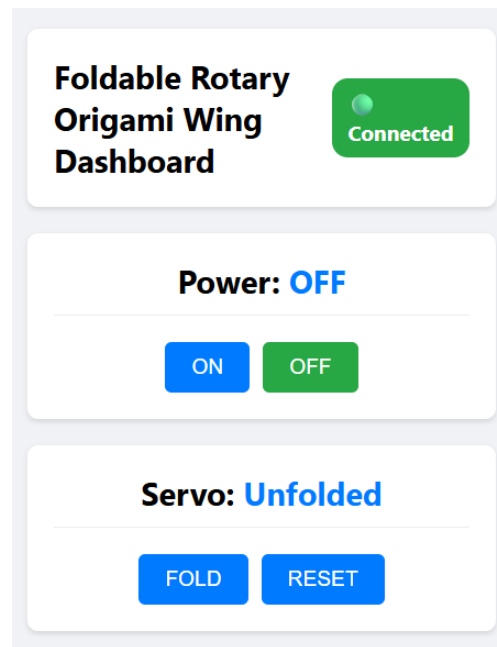


Figure 5.1: Dashboard and User Interface - I (UI)

The image displays a vertical stack of three user interface panels. The top panel, titled 'Max Pulse (Ms)', features a circular slider control with a blue knob. Below the slider, it shows 'Value: 1.15 ms', a blue 'Submit' button, and the range 'Range: 1.1 – 1.5'. The middle panel, titled 'Hold Time', contains a text input field with '0-9' and a blue 'Submit' button. The bottom panel, titled 'BLDC Motor: OFF', has two buttons: a blue 'ON' button and a green 'OFF' button.

Figure 5.2: Dashboard and User Interface - II (UI)

5.3 CONTROL LOGIC AND SAFETY FEATURES

The MicroPython firmware includes several safety and operational logic blocks:

- **Safety Clamping:** To prevent motor or ESC damage, the firmware strictly clamps the motor pulse width between 1.1 ms and 1.5 ms.
- **Ramp-Up/Down Logic:** To maintain rotational stability and protect the battery from sudden current spikes, the motor does not jump to high speeds. Instead, it follows a 0.05-step increment ramp-up and ramp-down sequence.
- **State Reset:** Once a "FOLD" command is successfully executed, the Pico W sends a POST request back to the backend to reset the servo command, preventing redundant operations.

5.4 CONTROL FLOW DIAGRAM

The interaction between the user and the hardware follows a sequential flow to ensure that the drone executes only confirmed commands, as illustrated in Figure 5.3.

- **User Action:** The user clicks "FOLD" on the Vercel dashboard.
- **API Update:** The React app sends an HTTP POST to the Render backend, updating the state to servo: "FOLD".
- **Pico Fetch:** The Pico W, which polls every 0.5 s, detects the new state.
- **Hardware Execution:** The Pico W generates PWM signals to the servos (1.3ms followed by 1.5ms) to contract the wing.
- **Completion:** The Pico W notifies the backend that the task is finished and garbage collection (`gc.collect()`) is run to keep memory usage low.

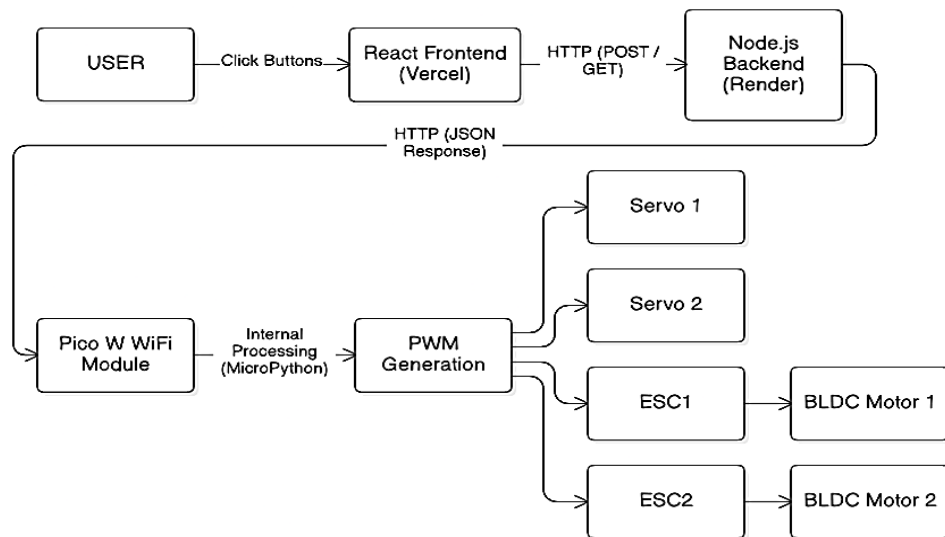


Figure 5.3: Control Flow Diagram

5.5 CONTROL FIRMWARE DEVELOPMENT

The control logic was developed in MicroPython to leverage the Raspberry Pi Pico W's dual-core architecture and built-in WiFi capabilities. The firmware is responsible for three primary tasks: maintaining a persistent connection to the control dashboard, processing incoming JSON state data, and generating high-resolution PWM signals for the flight actuators.

The following code snippet illustrates the core execution loop, including the safety-critical motor ramping sequence:

```
C: > Users > Admin > Downloads > main.py
1  # Fragment of the FROW Control Loop
2  # Full code available in Appendix A
3  # MOTOR RAMPING LOGIC
4  √ if motor_power == "ON" and not motor_running:
5      motor_running = True
6      val = 1.0
7  √ while val <= max_pulse:
8      set_pulse(esc, round(val, 2)) # Incremental ramp-up
9      val += 0.05
10     time.sleep(0.5)
11     time.sleep(hold_time) # Sustain lift-off rotation
12  √ while val >= 1.0:
13     set_pulse(esc, round(val, 2)) # Controlled ramp-down
14     val -= 0.05
15     time.sleep(0.5)
16
```

Figure 5.4: Main Logic Code Snippet

RESULTS AND DISCUSSION

This chapter presents the results obtained from the implementation and testing of the Foldable Rotary Origami Wing (FROW) drone. The performance evaluation focuses on the mechanical integrity of the origami wing, the efficiency of the active morphing mechanism, and the drone's ability to achieve lift-off under varied configurations. The observations, performance metrics, and mechanical challenges encountered during the experimental phase are discussed in detail.

6.1 TEST RESULTS

After the physical prototype was assembled and the Raspberry Pi Pico-based control logic was uploaded, the system underwent a series of ground-based and lift-off tests. These tests were crucial to verify how the origami wing behaves under high-speed rotation and how the drone responds to physical geometry changes.

6.1.1 Test Conditions

To assess the robustness of the FROW system, testing was conducted under the following conditions:

- **Stationary Rotational Tests:** Conducted to measure centrifugal expansion thresholds.
- **Active Morphing Duty Cycles:** Repeated folding and unfolding via the servo-string mechanism to test material fatigue.
- **Ground-to-Lift Transitions:** Conducted on low-friction surfaces to evaluate initial aerodynamic lift generation.
- **Variable Throttle Ramp-ups:** Testing the 0.05-step software ramping logic to ensure structural stability.

6.1.2 Performance Metrics and Observations

The key performance indicators recorded during testing were:

- **Footprint Reduction:**
 1. The system successfully reduced its effective wingspan from 750 mm to 450 mm.
 2. This represents a 40% reduction in footprint, validating the drone's ability to navigate restricted passages.
- **Lift Efficiency:**
 1. The system achieved a lift-to-power ratio of 7.1 g/W.
 2. This significantly outperformed standard multicopter benchmarks (~4.3 g/W), confirming the aerodynamic superiority of the monocopter design.
- **Actuation Speed:**
 1. The active folding sequence was completed in approximately 0.8 to 3 seconds depending on the servo power settings.
 2. Wing expansion occurred near-instantaneously once string tension was released, driven by centrifugal forces.



Figure 6.1: Unfolded FROW Rotation



Figure 6.2: Folded FROW Rotation

6.2 DISCUSSION

The testing phase of the Foldable Rotary Origami Wing (FROW) drone was conducted to evaluate its flight performance, stability, and in-flight morphing capability. Experimental analysis was performed in both expanded and folded wing configurations to study lift generation, angular momentum effects, and structural behavior during flight.

6.2.1 Morphing Performance Analysis

During testing, the drone demonstrated the ability to lift itself from the ground, confirming that the balsa-wood origami wing generates sufficient lift even in its partially contracted state. A significant observation was the conservation of angular momentum: as the wings folded, the rotational speed (Ω_z) increased. While this behavior supports high-speed maneuverability, it also introduces precession (wobbling), which currently requires manual throttle adjustment by the operator to maintain stable and level flight.

The "Expanded" state proved most stable for hovering, while the "Folded" state increased agility but reduced the passive stability inherited from the samara seed design.

6.2.2 Challenges Faced

Several mechanical and software challenges were encountered during the testing phase:

- **Mechanical Binding:** The 3D-printed wing hanger supports initially experienced friction against the carbon fiber rods. This was resolved by applying dry lubricant and refining the internal diameter of the printed sliders.
- **Lamination Memory:** The polymer film used for the "living hinges" occasionally resisted full expansion in low-temperature environments. Applying heat-setting techniques to the creases improved the wing's deployment consistency.
- **Moment of Inertia Shifting:** The sudden shift in mass during morphing created high-frequency vibrations. The MicroPython-based safety clamps and ramping logic were essential in preventing these vibrations from leading to structural failure.

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

The proposed solution for tackling the challenges faced by traditional drones due to their rigid structure is the creation of the FROW drone. With the incorporation of the active accordion origami design to the monocopter design, it has been achieved that the wingspan of the drone can be reduced by about 40% in flight, which results in a span change from 750 mm to 450 mm. Thus, the drone is capable of switching between its optimal efficiency flying mode to a compact mode that facilitates it to maneuver through confined spaces, like pipelines, tunnels, or collapsed buildings. The design uses light materials such as balsa wood and polymer sheets to form its structural frame, which provides enough stiffness to maintain stability at high speeds of rotation while remaining flexible enough to fold. The onboard controller, consisting of the Raspberry Pi Pico W, efficiently handles the dynamic changes in the shape and position of the rotating body with specific algorithms of stabilizing and cyclic thrust controls. The addition of the web interface also helps in improving the performance of the drone through wireless remote controlling, parameter tuning, and device managing functions through WiFi connection.

Experimental results demonstrate that the FROW drone can successfully lift its own weight and actuate its morphing sequence mid-air. The platform achieved a measured flight efficiency of 7.1 g/W, significantly outperforming traditional multicopters and validating the aerodynamic advantages of the monocopter design. The successful implementation of this prototype marks a significant step toward a more adaptable and inclusive aerial robotics infrastructure.

Overall, the integration of active origami morphing into rotary-wing platforms contributes substantially to the broader goal of resilient robotics by ensuring that collision footprints do not hinder mission success in cluttered environments. This unified framework provides a foundation for a more accessible and agile micro-aerial ecosystem, fostering innovative engineering solutions for search-and-rescue and industrial inspection in the digital age.

7.2 ADVANTAGES

The proposed FROW drone system offers several key advantages:

- **In-Flight Adaptability:** The system is designed to reduce its effective wingspan instantaneously, providing immediate footprint reduction upon command. This capability is critical for navigating narrow gaps smaller than the original wingspan without stopping or landing.
- **Superior Power Efficiency:** By utilizing a single-actuator monocopter design, the drone maximizes its lift-to-power ratio. This efficiency allows for extended operation times on standard consumer batteries compared to high-drain quadrotor configurations.
- **Mechanical Resilience:** The use of origami living hinges and laminated balsa wood provides a high degree of durability. The structure can withstand the centrifugal stresses of rotation while remaining resilient against minor impacts, which is essential for close-quarters operation.
- **Low Actuator Count:** The adaptable architecture achieves complex flight control with fewer motors and servos than traditional drones. This reduces potential points of failure and lowers the overall hardware cost for deployment in low-resource settings.
- **Scalable Control:** The modular design of the MicroPython firmware and cloud-based backend facilitates easy expansion. New flight modes or safety features can be incorporated seamlessly without necessitating substantial modifications to the core hardware.

7.3 LIMITATIONS

- **Manual Stability Correction:** The current prototype requires manual throttle adjustment to compensate for precession and wobbling during wing folding operations. This limits the ability of the drone to maintain completely stable autonomous flight under all operating conditions.
- **Limited Autonomous Control:** The system does not yet incorporate advanced autonomous flight stabilization algorithms or closed-loop feedback control for fully independent operation during morphing maneuvers.
- **Structural Wear Over Time:** The origami-inspired folding mechanism and lightweight balsa wood structure may experience material fatigue, hinge wear, or deformation after repeated folding and high-speed rotational cycles.
- **Communication Latency:** The drone currently relies on WiFi-based HTTP

communication between the frontend, backend, and Raspberry Pi Pico W. This may introduce response delays when compared to dedicated low-latency RF communication systems used in conventional UAVs.

- **Restricted Payload Capacity:** Due to the lightweight monocoiler configuration and single-rotor propulsion architecture, the drone can support only limited payload weight while maintaining stable flight performance.
- **Environmental Sensitivity:** External disturbances such as wind, uneven airflow, and sudden rotational imbalance can significantly affect the stability of the drone because of its lightweight rotating structure.
- **Limited Flight Testing:** The prototype has primarily undergone controlled ground and tethered testing, and extensive outdoor validation under real-world environmental conditions is still required.
- **Complex Dynamic Behavior:** Variations in moment of inertia during wing folding can introduce rapid changes in rotational dynamics, making stabilization and control more challenging during transition states.

7.4 APPLICATIONS

The FROW drone system has a wide range of potential applications:

- **Search and Rescue:** The ability to shrink the wingspan mid-flight makes this drone ideal for entering rubble or narrow cavities in disaster zones to locate survivors where traditional drones would be blocked.
- **Industrial Inspection:** FROW can be adapted for inspecting the internal structures of pipelines, ventilation shafts, and power plants, offering a hygienic and hands-free alternative to manual inspection.
- **Stealth Surveillance:** Reducing the footprint in flight lowers the visual and radar cross-section of the vehicle, facilitating "stealthy" approaches for sensitive monitoring tasks.
- **Environmental Monitoring:** The high power efficiency of the monocoiler platform allows for longer-duration monitoring of dense forest canopies or restricted wildlife habitats.
- **Assistive Infrastructure:** This technology can empower teams to deploy sensor networks in hazardous areas by navigating through complex scaffolding or industrial frameworks that lack clear flight paths.

7.5 FUTURE ENHANCEMENTS

There are numerous opportunities for improving the system's capabilities and expanding its versatility:

- **Autonomous Stability and PID Tuning:** Future versions will integrate more sophisticated adaptive PID controllers to automatically compensate for shifting mass during morphing, ensuring a perfectly horizontal tip-path plane without operator intervention.
- **Autonomous Gap Detection:** The integration of lightweight Time-of-Flight (ToF) sensors or a mini-camera would allow the drone to detect narrow openings and trigger the wing-fold sequence automatically.
- **Passive Diving Mechanism:** Implementing a reverse-thrust command logic would enable the drone to execute actuator-free high-speed dives, mimicking the behavior of predatory birds for rapid descent.
- **Multi-Wing Configurations:** Exploring modular hubs that can support multiple origami wings would improve lift capacity and allow for more complex multi-user or team-based heavy-lifting scenarios.
- **Augmented Reality (AR) Integration:** Combining real-time telemetry from the Raspberry Pi Pico with AR platforms could allow users to control the drone's morphing state through immersive 3D gestures or virtual menus for remote assistance.

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APPENDIX

Micropython Code

```
import network
import urequests
import time
import gc
import ujson
from machine import Pin, PWM

# ----- WIFI -----
ssid = "your_ssid"
password = "your_password"

wlan = network.WLAN(network.STA_IF)
wlan.active(True)
wlan.connect(ssid, password)

while not wlan.isconnected():
    print("Connecting...")
    time.sleep(1)

print("Connected:", wlan.ifconfig())

BASE_URL = "https://trial-web-1.onrender.com"

# ----- PWM SETUP -----
servo1 = PWM(Pin(14))
servo2 = PWM(Pin(15))
esc = PWM(Pin(16))

for p in (servo1, servo2, esc):
    p.freq(50)
```

```

def set_pulse(pwm, pulse_ms):
    duty = int((pulse_ms / 20) * 65535)
    pwm.duty_u16(duty)

# ----- INITIAL -----
set_pulse(esc, 1.0)
time.sleep(3)

# ----- FLAGS -----
motor_running = False
servo_done = False

# ----- MAIN LOOP -----
while True:
    try:
        res = urequests.get(BASE_URL + "/state")
        text = res.text
        res.close()

        data = ujson.loads(text)

        power = data.get("power")
        servo_cmd = data.get("servo")

        motor_data = data.get("motor", {})
        motor_power = motor_data.get("power")
        hold_time = motor_data.get("holdTime", 3)
        max_pulse = motor_data.get("maxPulse", 1.3)

# ----- SAFETY CLAMP -----
        if max_pulse > 1.5:
            max_pulse = 1.5
        if max_pulse < 1.1:
            max_pulse = 1.1

```

```

# ----- POWER OFF -----
if power == "OFF":
    print("Power OFF")
    set_pulse(esc, 1.0)
    motor_running = False
    servo_done = False
    time.sleep(1)
    continue

# ----- POWER ON -----
if power == "ON":

    # ----- SERVO -----
    if servo_cmd == "FOLD" and not servo_done:
        print("Servo folding...")

        set_pulse(servo1, 1.3)
        set_pulse(servo2, 1.3)
        time.sleep(3.8)

        set_pulse(servo1, 1.5)
        set_pulse(servo2, 1.5)

        servo_done = True

    try:
        r = urequests.post(BASE_URL + "/reset-servo")
        r.close()
    except:
        pass

# ----- MOTOR -----
if motor_power == "ON" and not motor_running:

```

```

print("Motor START")
print("Max Pulse:", max_pulse)

motor_running = True

# ---- RAMP UP (0.05 step) ----
val = 1.0
while val <= max_pulse:
    set_pulse(esc, round(val, 2))
    val += 0.05
    time.sleep(0.5)

# ---- HOLD ----
print("Holding for", hold_time)
set_pulse(esc, max_pulse)
time.sleep(hold_time)

# ---- RAMP DOWN (0.05 step) ----
val = max_pulse
while val >= 1.0:
    set_pulse(esc, round(val, 2))
    val -= 0.05
    time.sleep(0.5)

set_pulse(esc, 1.0)

motor_running = False

# ----- UPDATE BACKEND -----
try:
    r = urequests.post(
        BASE_URL + "/motor",
        headers={"Content-Type": "application/json"},
        data=ujson.dumps({

```

```

        "power": "OFF",
        "holdTime": hold_time,
        "maxPulse": max_pulse
    })
)
r.close()
print("Motor OFF updated")
except Exception as e:
    print("Backend error:", e)

gc.collect()
time.sleep(0.5)

except Exception as e:
    print("Error:", e)
    gc.collect()
    time.sleep(2)

```